THE TWO DEFINITIONS OF COMPLETE INTERSECTION AGREE FOR FINITE TYPE ALGEBRAS OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written. Declared dependency: lemma-quotient-regular-ring is not carded; recorded via external_references.

Allowed standard prerequisites (editorial).

- Local CI criterion over a regular ring. lemma-quotient-regular-ring (same chapter, named; NOT carded – statement carries an unfinished item). The dpa-side of the comparison; declared as an external reference, not re-proved.
- Algebra-side local and global criteria. algebra-lemma-lci-local, algebra-lemma-lci-global (cross-chapter, named). The Algebra-side of the comparison and the global step.

Source reference. Original reference 00S9: focused-seeds/geometry/S9-dpa-check-lci-agrees/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00SD: focused-seeds/geometry/S9-dpa-check-lci-agrees/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00SH: focused-seeds/geometry/S9-dpa-check-lci-agrees/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00SF: focused-seeds/geometry/S9-dpa-check-lci-agrees/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00OQ: focused-seeds/geometry/S9-dpa-check-lci-agrees/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 09Q3: sources/stacks-dpa/pinned-original-source.tex, source lines 2137–2137. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09Q1: sources/stacks-dpa/pinned-original-source.tex, source lines 2010–2010. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a finite type algebra over a field k.*

(1) *for a prime $\mathfrak{q} \subset S$ the local ring $S_{\mathfrak{q}}$ is a complete intersection in the sense of Algebra, Definition 00SD if and only if $S_{\mathfrak{q}}$ is a complete intersection in the sense of Definition 09Q3, and*
(2) *S is a local complete intersection in the sense of Algebra, Definition 00S9 if and only if S is a local complete intersection in the sense of Definition 09Q3.*

Proof. Proof of (1). Let $k[x_1, \ldots, x_n] \to S$ be a surjection. Let $\mathfrak{p} \subset k[x_1, \ldots, x_n]$ be the prime ideal corresponding to $\mathfrak{q}$. Let $I \subset k[x_1, \ldots, x_n]$ be the kernel of our surjection. Note that $k[x_1, \ldots, x_n]_{\mathfrak{p}} \to S_{\mathfrak{q}}$ is surjective with kernel $I_{\mathfrak{p}}$. Observe that $k[x_1, \ldots, x_n]$ is a regular ring by Algebra, Proposition 00OQ. Hence the equivalence of the two notions in (1) follows by combining Lemma 09Q1 with Algebra, Lemma 00SF.

Having proved (1) the equivalence in (2) follows from the definition and Algebra, Lemma 00SH. □